

AZ-305 AI Study Guide

Weekly Learning Plan & Checklist — John's Study Plan

W1 Azure AI Services Foundations			🕒 6-8 hrs	0/8 ▾
Understand the AI service landscape on Azure				
☐		Read: AZ-305 Official Study Guide (April 2026 update) learn.microsoft.com — Study Guide	30 min	
☐		Module: AZ-305 Exam Overview & Audience Profile learn.microsoft.com — Exam Page	20 min	
☐		Read: What are Azure AI Services? learn.microsoft.com — AI Services	45 min	
☐		Module: Introduction to Azure AI Services learn.microsoft.com — AI Fundamentals	1 hr	
☐		Read: Azure OpenAI Service Overview learn.microsoft.com — OpenAI Service	45 min	
☐		Module: Intro to Azure OpenAI Service learn.microsoft.com — Explore OpenAI	1 hr	
☐		Read: Azure AI Foundry Overview learn.microsoft.com — AI Foundry	30 min	
☐		Lab: Take the free AZ-305 Practice Assessment learn.microsoft.com — Practice Assessment	1 hr	

W2

Designing Data Storage for AI

Choose the right data layer for AI workloads

 7–9 hrs0/8 

Learning Path: Design Data Storage Solutions (AZ-305)

2 hr

learn.microsoft.com — Data Storage Path



Read: Azure Cosmos DB Introduction

45 min

learn.microsoft.com — Cosmos DB



Read: Cosmos DB for AI & Vector Search

30 min

learn.microsoft.com — Vector Search



Read: Azure AI Search Overview

45 min

learn.microsoft.com — AI Search



Module: Implement Azure AI Search

1 hr

learn.microsoft.com — Knowledge Store



Read: Azure Data Lake Storage Gen2 Overview

30 min

learn.microsoft.com — Data Lake Gen2



Read: Azure Synapse Analytics for AI/ML

45 min

learn.microsoft.com — Synapse Analytics



Lab: Design a non-relational data storage solution

1 hr

learn.microsoft.com — Non-relational Data

W3

Azure Machine Learning Architecture

Architect robust ML pipelines and deployments

🕒 6–8 hrs

0/8 ▾

**Read: Azure Machine Learning Overview**

45 min

[learn.microsoft.com — Azure ML Overview](https://learn.microsoft.com/Azure/ML/Overview)**Module: Design an Azure ML Solution**

1 hr

[learn.microsoft.com — Design ML Solution](https://learn.microsoft.com/Azure/ML/Design-ML-Solution)**Read: Choose Azure ML Compute Targets**

30 min

[learn.microsoft.com — Compute Targets](https://learn.microsoft.com/Azure/ML/Compute-Targets)**Read: MLOps with Azure Machine Learning**

45 min

[learn.microsoft.com — MLOps](https://learn.microsoft.com/Azure/ML/MLOps)**Module: Introduction to MLOps on Azure**

1.5 hr

[learn.microsoft.com — MLOps Path](https://learn.microsoft.com/Azure/ML/MLOps-Path)**Read: Deploy ML Models with Managed Endpoints**

30 min

[learn.microsoft.com — Managed Endpoints](https://learn.microsoft.com/Azure/ML/Managed-Endpoints)**Read: Azure Databricks for ML Workloads**

30 min

[learn.microsoft.com — Databricks ML](https://learn.microsoft.com/Azure/ML/Databricks-ML)**Lab: Azure ML Workspace Hands-On (MS Learn Sandbox)**

1 hr

[learn.microsoft.com — ML Workspace Lab](https://learn.microsoft.com/Azure/ML/ML-Workspace-Lab)

W4

Generative AI & RAG Architecture

Design Azure OpenAI and RAG solutions

🕒 6–8 hrs

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| ☐ |  Read: Retrieval-Augmented Generation (RAG) on Azure
learn.microsoft.com — RAG Overview | 45 min |
| ☐ |  Module: Build Generative AI Solutions with Azure OpenAI
learn.microsoft.com — Gen AI Path | 2 hr |
| ☐ |  Read: Azure OpenAI On Your Data (RAG Pattern)
learn.microsoft.com — On Your Data | 45 min |
| ☐ |  Read: Azure AI Content Safety Overview
learn.microsoft.com — Content Safety | 30 min |
| ☐ |  Module: Architect Intelligent Apps with Azure AI Foundry
learn.microsoft.com — AI Foundry | 45 min |
| ☐ |  Read: Semantic Kernel for Azure AI Orchestration
learn.microsoft.com — Semantic Kernel | 30 min |
| ☐ |  Read: Azure API Management for AI APIs
learn.microsoft.com — API Management | 30 min |
| ☐ |  Lab: Chat with your data using Azure OpenAI
learn.microsoft.com — Use Own Data | 1 hr |

W5

Identity, Security & Governance for AI

Secure and govern AI workloads on Azure

🕒 5-7 hrs

0/8 ▼

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| ☐ |  Learning Path: Design Identity, Governance & Monitoring (AZ-305) 1.5 hr
learn.microsoft.com — Identity & Governance |
| ☐ |  Read: Managed Identities for Azure Resources 30 min
learn.microsoft.com — Managed Identities |
| ☐ |  Read: Azure Key Vault for AI Secrets & Keys 30 min
learn.microsoft.com — Key Vault |
| ☐ |  Read: Azure Policy for AI Service Governance 30 min
learn.microsoft.com — Azure Policy |
| ☐ |  Read: Responsible AI Principles on Azure 30 min
learn.microsoft.com — Responsible AI |
| ☐ |  Module: Network Security for Azure AI Services 45 min
learn.microsoft.com — VNet Security |
| ☐ |  Read: Private Endpoints for Azure AI Services 30 min
learn.microsoft.com — Private Endpoints |
| ☐ |  Lab: Microsoft Purview for AI Data Governance 45 min
learn.microsoft.com — Purview Governance |

W6

Monitoring, Integration & Exam Prep

Operationalize AI solutions and lock in exam readiness

🕒 6–8 hrs

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Learning Path: Design Business Continuity Solutions (AZ-305)

1.5 hr

[learn.microsoft.com — Business Continuity](https://learn.microsoft.com/azure/business-continuity)

Read: Azure Monitor for AI Workloads

30 min

[learn.microsoft.com — Azure Monitor](https://learn.microsoft.com/azure/monitor)

Read: Azure Well-Architected Framework (AI Pillar)

1 hr

[learn.microsoft.com — Well-Architected](https://learn.microsoft.com/azure/well-architected)

Read: Azure Architecture Center – AI + ML Patterns

45 min

[learn.microsoft.com — AI Architecture](https://learn.microsoft.com/azure/architecture)

Module: Design Infrastructure Solutions (AZ-305)

1.5 hr

[learn.microsoft.com — Infra Solutions](https://learn.microsoft.com/azure/infrastructure)

Read: Azure Event Grid for AI Event-Driven Pipelines

30 min

[learn.microsoft.com — Event Grid](https://learn.microsoft.com/azure/event-grid)

Practice: AZ-305 Practice Assessment (Full Run)

1 hr

[learn.microsoft.com — Practice Assessment](https://learn.microsoft.com/azure/practice-assessment)

Read: Schedule the AZ-305 Exam

15 min

[learn.microsoft.com — Schedule Exam](https://learn.microsoft.com/azure/schedule-exam)